Note on the Levi-Civita planar regularization of the Three-Body problem

di Riccardo Daluiso

1

Let $\Sigma' = Gxy$ be the inertial frame with the origin in the center of mass of the three points. In this system let $\underline{r}_\nu = (x_\nu, y_\nu)$ be the position vectors of the relative distances subjected to the constraint

$$\sum_{\nu=1}^{3} x_\nu = 0, \qquad \sum_{\nu=1}^{3} y_\nu = 0. \tag{1}$$

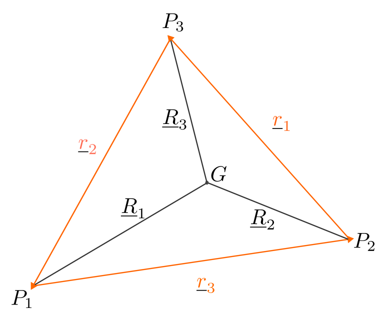

Figure 1: *Mutual distances of the three body in the plane. The origin is in the center of mass G.*

We implement the transformation for $\nu = 1, 2, 3$ defined by

$$x_\nu + iy_\nu = (\xi_\nu + i\eta_\nu)^2, \tag{2}$$

or equivalently

$$\begin{cases} x_\nu = \xi_\nu^2 - \eta_\nu^2, \\ y_\nu = 2\xi_\nu\eta_\nu. \end{cases} \tag{3}$$

If $\rho_\nu^2 = \xi_\nu^2 + \eta_\nu^2$, we have

$$r_\nu = \rho_\nu^2$$

and the expression for U in terms of the new variables becomes

$$U = \gamma M \sum_{\nu=1}^{3} \frac{m_\nu^*}{\rho_\nu^2}. \tag{4}$$

The relations (1) can be expressed in terms of the new variables, thus taking the form

$$\sum_{\nu=1}^{3}\left(\xi_\nu^2-\eta_\nu^2\right)=0, \qquad \sum_{\nu=1}^{3}\xi_\nu\eta_\nu=0. \tag{5}$$

The costraints above define a $4-$dimensional manifold embedded in $\mathbb{R}^6$. In order to prove this, we apply the preimage theorem to the function

$$\begin{aligned} &F:\mathbb{R}^6\to\mathbb{R}^2,\\ &F(\xi,\eta)=\Big(\sum_{\nu=1}^{3}\left(\xi_\nu^2-\eta_\nu^2\right),\ \sum_{\nu=1}^{3}\xi_\nu\eta_\nu\Big). \end{aligned} \tag{6}$$

It is sufficient to show that the matrix

$$J=\begin{pmatrix} 2\xi_1 & 2\xi_2 & 2\xi_3 & -2\eta_1 & -2\eta_2 & -2\eta_3\\ \eta_1 & \eta_2 & \eta_3 & \xi_1 & \xi_2 & \xi_3 \end{pmatrix}$$

has rank 2. Indeed

$$det(J^TJ)=4\left(\rho_1^2+\rho_2^2+\rho_3^2\right)$$

is equal to zero if and only if all the variables ξ_ν,η_ν vanish simultaneously.
From now on we will suppose that the collision arises from P_2 and P_3. Let $r:=r_1$ and $\rho:=\rho_1$ so that $r,\rho\to 0$. By the Implicit function theorem we can use as coordinates $\xi_1,\eta_1,\xi_2,\eta_2$, or $\xi_1,\eta_1,\xi_3,\eta_3$. Instead it is not possible to explicit ξ_1,η_1 as a function of $\xi_2,\eta_2,\xi_3,\eta_3$, as the intuition suggests. In this case, the relevant minor should be

$$\begin{vmatrix} 2\xi_1 & -2\eta_1\\ \eta_1 & \xi_1 \end{vmatrix}=\rho,$$

which is zero because $\rho=\sqrt{r}$.

Consider the restriction of (2) to the $\nu-$th body

$$\varphi(\xi_\nu,\eta_\nu)=\left(\xi_\nu^2-\eta_\nu^2,\ \ 2\xi_\nu\eta_\nu\right).$$

The associated momenta transform as

$$(p_{x_\nu},p_{y_\nu})=\left(\frac{\partial\varphi}{\partial(\xi_\nu,\eta_\nu)}\right)^{-T}\begin{pmatrix} p_{\xi_\nu}\\ p_{\eta_\nu}\end{pmatrix},$$

where in this case

$$\left(\frac{\partial\varphi}{\partial(\xi_\nu,\eta_\nu)}\right)^{-T}=\frac{1}{2(\xi_\nu^2+\eta_\nu^2)}\begin{pmatrix} \xi_\nu & -\eta_\nu\\ \eta_\nu & \xi_\nu. \end{pmatrix}$$

Therefore, we obtain the following relations between the old and the new momenta:

$$\begin{cases} p_{x_\nu}=\frac{1}{2\rho_\nu^2}\left(\xi_\nu p_{\xi_\nu}-\eta_\nu p_{\eta_\nu}\right),\\ p_{y_\nu}=\frac{1}{2\rho_\nu^2}\left(\eta_\nu p_{\xi_\nu}+\xi_\nu p_{\eta_\nu}\right), \end{cases} \qquad \nu=1,2,3. \tag{7}$$

In vector form this transformation reads

$$\begin{pmatrix} p_{x_\nu} \\ p_{y_\nu} \end{pmatrix} = \frac{1}{2\rho_v^2} \begin{pmatrix} \xi_\nu & -\eta_\nu \\ \eta_\nu & \xi_\nu \end{pmatrix} \begin{pmatrix} p_{\xi_\nu} \\ p_{\eta_\nu} \end{pmatrix}, \tag{8}$$

and it is easily inverted as

$$\begin{pmatrix} p_{\xi_\nu} \\ p_{\eta_\nu} \end{pmatrix} = 2 \begin{pmatrix} \xi_\nu & \eta_\nu \\ -\eta_\nu & \xi_\nu \end{pmatrix} \begin{pmatrix} p_{x_\nu} \\ p_{y_\nu} \end{pmatrix}. \tag{9}$$

Let $\underline{\hat{p}}_\nu := (p_{\xi_\nu}, p_{\eta_\nu})$ and $\underline{p}_\nu := (p_{x_\nu}, p_{y_\nu})$. Then,

$$\hat{p}_\nu^2 := p_{\xi_\nu}^2 + p_{\eta_\nu}^2 = 4\rho_\nu^2(p_{x_\nu}^2 + p_{y_\nu}^2). \tag{10}$$

We now prove that $\hat{p}_\nu$ remains finite at collision for each $\nu = 1, 2, 3$. For this purpose, it is important to relate the relative momenta to the absolute ones. So let $\underline{v}_\nu = \underline{\dot{r}}_\nu$. The kinetic energy in these coordinate reads

$$T = \frac{1}{2} \sum_{\nu=1}^{3} m_\nu^* v_\nu^2, \tag{11}$$

where

$$m_1^* = \frac{m_2 m_3}{M}, \quad m_2^* = \frac{m_1 m_3}{M}, \quad m_3^* = \frac{m_1 m_2}{M}.$$

Then

$$\underline{p}_\nu = \frac{\partial T}{\partial \underline{\dot{r}}_\nu} = m_\nu^* \underline{\dot{r}}_\nu. \tag{12}$$

For notational convenience, we use q_ν to denote one of the coordinates of the $\nu-$th body in the center of mass system of reference, and $V_\nu = \dot{q}_\nu$. Moreover, let $\underline{P}_\nu$ the absolute momenta of the $\nu-$th mass point, we have

$$\begin{aligned}
\underline{p}_1 &= m_1^* \underline{\dot{r}_1} = \frac{m_2 m_3}{M} \left(\underline{V_3} - \underline{V_2}\right) = \frac{1}{M} \left(m_2 \underline{P}_3 - m_3 \underline{P}_2\right), \\
\underline{p}_2 &= m_2^* \underline{\dot{r}_2} = \frac{m_1 m_3}{M} \left(\underline{V_1} - \underline{V_3}\right) = \frac{1}{M} \left(m_3 \underline{P}_1 - m_1 \underline{P}_3\right), \\
\underline{p}_3 &= m_3^* \underline{\dot{r}_3} = \frac{m_1 m_2}{M} \left(\underline{V_2} - \underline{V_1}\right) = \frac{1}{M} \left(m_1 \underline{P}_2 - m_2 \underline{P}_1\right).
\end{aligned} \tag{13}$$

From the definition of U,

$$\lim_{t \to t^*} rU = \gamma m_2 m_3, \tag{14}$$

therefore

$$U = O\left(\frac{1}{r}\right) \quad \text{as } \ t \to t^*.$$

From the energy integral

$$\frac{1}{2} \sum_{\nu=1}^{3} m_\nu V_\nu^2 = T = U + h,$$

using (14), and since rh tends to 0 at collision,

$$r\sum_{\nu=1}^{3} m_\nu V_\nu^2 = 2r\left(U+h\right) \xrightarrow[t\to t^*]{} 2\gamma m_2 m_3. \tag{15}$$

In particular rV_ν^2, rq_ν^2, $\sqrt{r}q_\nu$ for $\nu = 1,2,3$ remain finite at collision between P_2 and P_3. We want to estimate the asymptotic behaviour of V_2, V_3 at collision. Differentiation the center of mass relation $m_1q_1 + m_2q_2 + m_3q_3 = 0$ gives

$$\begin{gathered} rm_2^2\dot{q}_2^2 = \left(\sqrt{r}m_1\dot{q}_1 + \sqrt{r}m_3\dot{q}_3\right)^2, \\ r\left(m_2\dot{q}_2\right)^2 - r\left(m_3\dot{q}_3\right)^2 = m_1\sqrt{r}\left\{m_1\sqrt{r}\dot{q}_1^2 + 2m_3\dot{q}_1\left(\sqrt{r}\dot{q}_3\right)\right\}, \end{gathered}$$

where in the last equation each term on the left side is bounded, $\sqrt{r}\dot{q}_3$ is bounded and $\dot{q}_1$ is finite since P_1 does not collide. Then, as $t \to t^*$

$$\begin{gathered} r\left(m_2\dot{q}_2\right)^2 - r\left(m_3\dot{q}_3\right)^2 \longrightarrow 0, \\ r\left(m_2V_2\right)^2 - r\left(m_3V_3\right)^2 \longrightarrow 0. \end{gathered} \tag{16}$$

We multiply (15) by m_2 to obtain

$$rm_2m_1V_1^2 + rm_2^2V_2^2 + rm_2m_3V_3^2 \xrightarrow[t\to t^*]{} 2\gamma m_2^2 m_3.$$

Since $rm_2m_1V_1^2$ goes to 0, we have

$$rm_2^2V_2^2 + rm_2m_3V_3^2 \xrightarrow[t\to t^*]{} 2\gamma m_2^2 m_3,$$

and using relation (16)

$$rV_3^2\left(m_3^2 + m_2m_3\right) \xrightarrow[t\to t^*]{} 2\gamma m_2^2 m_3,$$

that is

$$rV_3^2 \xrightarrow[t\to t^*]{} \gamma\frac{2m_2^2}{m_2+m_3}. \tag{17}$$

Similarly, we obtain the limit for rV_2^2 as

$$rV_2^2 \xrightarrow[t\to t^*]{} \gamma\frac{2m_3^2}{m_2+m_3}, \tag{18}$$

and this determines the asymptotic behaviour of V_2 and V_3 as $t \to t^*$. In particular, $V_{2,3} = O\left(1/\sqrt{r}\right)$.

Therefore

$$\sqrt{r}\,\frac{m_2m_3}{M}(\underline{V}_3 - \underline{V}_2) \longrightarrow \frac{m_2m_3}{M}\sqrt{2\gamma(m_2+m_3)}\,\underline{e}.$$

where $\underline{e}$ is a unit vector in the collision direction. Taking norms yields

$$r\left|\frac{m_2m_3}{M}(\underline{V}_3 - \underline{V}_2)\right|^2 \longrightarrow \frac{2\gamma\, m_2^2m_3^2(m_2+m_3)}{M^2}.$$

Then

$$p_1 = O\left(\frac{1}{\sqrt{r}}\right) = O\left(\frac{1}{\rho}\right) \tag{19}$$

and from (10)

$$\hat{p}_1 = 2\rho\, O\left(\frac{1}{\rho}\right) = O(1) \tag{20}$$

remains finite, as we claimed. The same holds for $\hat{p}_2$ and $\hat{p}_3$.